

# Yash Sanjay Bhalgat

Senior Research Scientist, Google DeepMind

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## WORK EXPERIENCE

**Google DeepMind**, Senior Research Scientist – Gemini Omni (Veo) [Nov '25 - Present]

- Core contributor in the “pretraining” and “video editing” pillars of the Gemini Omni video generation model [Blog].
- Lead captain** for Omni v1.5 – built the code infrastructure for Torsopatch pretraining, ran several model training ladders to incorporate critical capabilities into the model, and launched the final model training run for Omni v1.5.
- Co-led 3D/camera capabilities in Omni v1 – built a camera-captioning pipeline with significantly stronger text-to-camera control than Veo 3.1; also, shipped novel “Camera Reshoot” and “Camera Motion Copy” capabilities.
- Built the “Variable-Length Captioning” pipeline for video editing, enabling segment-level granularity for edits in Omni.
- Awarded 3 peer bonuses to recognize exceptional work done on 3 different projects.

**Meta Reality Labs**, Research Scientist Intern [Apr '25 - Sep '25]

- Built 4D motion-conditioned video generation world models for robotics.

**Multiple startups**, Part-time AI Consultant [Feb '23 - Mar '25]

- SoftEye Inc** (acquired by TDK Corp): Developing real-time 3D vision algorithms for augmented reality on smart glasses.
- GoBubble Ltd**: Finetuned and deployed Large Language Model (LLM) models to moderate multimodal data online.
- Togal.AI**: Building multimodal solutions for detecting and understanding project features on architectural plans/drawings.

**Qualcomm AI Research** | Senior Machine Learning Researcher [Nov '20 - Jul '21]  
Machine Learning Researcher [Jun '19 - Oct '20]

- Spearheaded the ultra-low resource always-on vision project from model design, quantization to final hardware mapping
- Filed 12 inventions in 2020-21 of which 6 ideas have been filed for patent protection. Notable works on 3D hand-pose estimation [DIR-Net], low-bit quantization [LSQ+, QKD], structured [StructConv] and unstructured [LTP] pruning
- Led Qualcomm’s team in the MicroNet Challenge at NeurIPS 2019, and achieved 3rd position in ImageNet track [Code]
- Managed/mentored interns - Jangho Kim and John Yang (PhD @ SNU) with contributions to the AR/VR project

**Voxel51, Inc.**, Computer Vision & Machine Learning Engineer [Feb '19 - May '19]

- Researched and developed production pipelines for real-time vehicle tracking for querying on large-scale video databases

## EDUCATION

**University of Oxford** Oct '21 - Nov '25

PhD (DPhil), Computer Vision and Machine Learning @ Visual Geometry Group (VGG)

**Thesis**: From video to virtual: object-centric 3D scene understanding from videos

**Advisors**: Prof. Andrew Zisserman, Prof. Andrea Vedaldi, Dr. João Henriques, Dr. Iro Laina

**University of Michigan, Ann Arbor** Sep '17 - Dec '18

Masters, Computer Science and Engineering

**Indian Institute of Technology, Bombay** Jul '13 - May '17

B.Tech. (with Honors) in Electrical Engineering with Minor in Computer Science

**Thesis**: Scattering Wavelets Network based Latent Fingerprint Enhancement, **Advisor**: Prof. Vikram Gadre

Undergraduate Research Award (URA 02) for exceptional Bachelors Thesis

## SELECTED PUBLICATIONS

**Conference Publications** (Full list: [Google scholar](https://scholar.google.com/citations?user=UWUWUWUWUW)) \* equal contribution

- Do 3D Large Language Models Really Understand 3D Spatial Relationships?. [Paper] [Project]  
ICLR, 2026 (**Best paper honorable award, 3D-LLM/VLA Workshop, CVPR 2026**). X. Ma, T. Sun, S. Chen, Yash Bhalgat, J. Gu, A. Chang, I. Armeni, I. Laina, S. Peng, V. Prisacariu.
- Jamais Vu: Exposing the Generalization Gap in Supervised Semantic Correspondence. [Paper] [Code]  
NeurIPS, 2025. Octave Mariotti, Zhipeng Du, Yash Bhalgat, Oisín Mac Aodha, Hakan Bilen.
- Reflecting Reality: Enabling Diffusion Models to Produce Faithful Mirror Reflections. [Paper] [Code] [Dataset]  
3DV, 2025. A. Dhiman\*, M. Shah\*, R. Parihar, Yash Bhalgat, L. Boregowda, R Venkatesh Babu.
- GS-CPR: Efficient Camera Pose Refinement via 3D Gaussian Splatting. [Paper] [Code]  
ICLR, 2025. C. Liu, S. Chen, Yash Bhalgat, S. Hu, M. Cheng, Z. Wang, V. Prisacariu, T. Braud.

8. 3D-Aware Instance Segmentation and Tracking in Egocentric Videos. [Paper]  
**ACCV**, 2024. *Yash Bhalgat\**, Vadim Tschernezki\*, Iro Laina, João Henriques, Andrea Vedaldi, Andrew Zisserman.
7. N2F2: Hierarchical Scene Understanding with Nested Neural Feature Fields. [Paper]  
**ECCV**, 2024. *Yash Bhalgat*, Iro Laina, João Henriques, Andrew Zisserman, Andrea Vedaldi.
6. SiLVR: Scalable Lidar-Visual Reconstruction with Neural Radiance Fields for Robotic Inspection. [Paper]  
**ICRA**, 2024. Yifu Tao, *Yash Bhalgat*, Lanke Frank Tarimo Fu, Matias Mattamala, Nived Chebrolu, Maurice Fallon.
5. Neural Refinement for Absolute Pose Regression with Feature Synthesis. [Paper]  
**CVPR**, 2024. Shuai Chen, *Yash Bhalgat*, Xinghui Li, Jiawang Bian, Kejie Li, Zirui Wang, Victor Adrian Prisacariu.
4. Contrastive Lift: 3D Object Instance Segmentation by Slow-Fast Contrastive Fusion. [Paper][Code]  
**NeurIPS**, 2023 (**Spotlight**). *Yash Bhalgat*, Iro Laina, João Henriques, Andrea Vedaldi, Andrew Zisserman.
3. A Light Touch Approach to Teaching Transformers Multi-view Geometry. [Paper]  
**CVPR**, 2023. *Yash Bhalgat*, João Henriques, Andrew Zisserman.
2. A Prompt Array Keeps the Bias Away: Debiasing Vision-Language Models with Adversarial Learning. [Paper]  
**AAACL-IJCNLP**, 2022. Hugo Berg, Siobhan Hall, *Yash Bhalgat*, Wonsuk Yang, Hannah Kirk, A. Shtedritski, M. Bain.
1. Structured Convolutions for Efficient Neural Network Design. [Paper]  
**NeurIPS**, 2020. *Yash Bhalgat*, Yizhe Zhang, Jamie Lin, Fatih Porikli.

## SELECTED INTERNSHIPS & PROJECTS

- [Project] **NeurIPS '19 MicroNet challenge - 3rd place, ImageNet track** [Code] [Jul '19 - Oct '19]
  - Designed fast evolutionary mixed precision quantization: 8x compression EfficientNet-B0/MixNet-S, <1% accuracy loss
- [Internship] **IBM Almaden Research Center, Mentor - Zhe Liu, Pritam Gundecha** [Summer '18]
  - Proposed teacher-student learning paradigm for task-agnostic classification in presence of label noise in train data [Paper]
- [Internship] **IFP Energies nouvelles, Paris, Mentor - Laurent Duval** [Summer '17]
  - Real-time sparsity-based deformation-invariant feature extraction for large geophysical images (ICASSP 2018) [Paper]
- [Internship] **IBM Research, Bangalore, Mentor - Vikas Raykar** [Summer '16]
  - Joint multi-modal representations for e-commerce catalog search by visual attributes *without* manual tagging

## SKILLS

|                   |                                                                                                       |
|-------------------|-------------------------------------------------------------------------------------------------------|
| <b>Languages</b>  | Python (proficient), C++ (moderate), Julia, MATLAB, Verilog, Bash, $\LaTeX$                           |
| <b>Frameworks</b> | PyTorch, Jax (proficient), CUDA, LangChain, DSPy, 🤖 Diffusers, TensorFlow & Keras (basic), git, slurm |

## TEACHING EXPERIENCE

|                                                                      |                                                                   |                         |
|----------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------|
| <b>University of Oxford,</b><br><i>Teaching Assistant</i>            | Computer Vision, w/ <i>Profs Andrea Vedaldi, Andrew Zisserman</i> | [Hillary '22, '24, '25] |
|                                                                      | Computer Graphics, with <i>Dr. Jassim Happa, Stuart Golodetz</i>  | [Hillary '22]           |
|                                                                      | Artificial Intelligence, with <i>Prof. Bernardo Cuenca Grau</i>   | [Hillary '22]           |
| <b>University of Michigan,</b><br><i>Graduate Student Instructor</i> | Computational Data Science, with <i>Prof. Raj Nadakuditi</i>      | [Fall '18]              |
|                                                                      | Introduction to Logic Design, with <i>Prof. Matthew Smith</i>     | [Winter '18]            |
| <b>IIT Bombay,</b><br><i>Teaching Assistant</i>                      | Wavelets, with <i>Prof. Vikram Gadre</i>                          | [Fall '16, Winter '17]  |
|                                                                      | Quantum Mechanics and Applications, with <i>Prof. Siva Prasad</i> | [Fall '14, Winter '15]  |

## SCHOLASTIC ACHIEVEMENTS

- Undergraduate Research Award (URA 02) for exceptional work during Bachelors Thesis at IIT Bombay
- Cargill Global Scholarship 2014-15 and 2015-16 for excellence in leadership and academic skills
- All India Rank **12** in IITJEE-Mains exam among 1 million candidates
- All India Rank **155** in IITJEE-Advanced exam among 150,000 candidates
- All India Rank **60** in KVPY Scholarship exam by Govt. of India among 0.2 million candidates
- Selected among National Top 30 (for OCSC camp) for International Astronomy Olympiad '13
- Selected among top 300 participants of India to compete in **all three national olympiads**: INPhO (Indian National Physics Olympiad), INChO (Chemistry), INAO (Astronomy)
- Visharad Degree (*equivalent to Bachelors in Music*) in Indian Classical Music for playing Tabla